Faces or Facts?

Experimental Evidence on Time Pressure, Information Costs, and Gender Bias in Electoral Choice

Alejandra Marín Díaz
Postdoctoral Researcher
O'Higgins University
alejandra.marin@postdoc.uoh.cl

Andrea Canales
Central Bank, Chile
acanalesg@.bcentral.cl

Héctor Bahamonde
Senior Researcher
University of Turku
hector.bahamonde@utu.fi

Work in progress — 19 October 2026

Download the latest version

Abstract

Voters often see a candidate's face before learning about the candidate's political positions. Building on evidence that rapid visual impressions predict electoral outcomes, we examine how time pressure and information costs shape political learning and whether these constraints produce distinct patterns of scrutiny for women candidates. We conduct a preregistered three-arm online visual conjoint experiment using photographs and political information drawn from Colombian electoral data. Respondents are randomly assigned to make these choices under a nine-second deadline, an effort cost for accessing political information, or an unconstrained baseline. Candidate pairings, task order, and left-right placement are randomized within respondents. The experiment measures recorded candidate selection, candidate-specific information acquisition, and decision time. The design targets causal effects of the decision environment and of assigning complete candidate profiles. Reported analyses use eligible completed interviews and are interpreted as completion-conditioned contrasts because completion may itself respond to treatment. Because the attributes of real candidates are not independently randomized, analyses of gender, ideology, and appearance characterize heterogeneity across observed profile bundles rather than attribute-level average marginal component effects.

1 Faces arrive before facts

Most political choices begin with a sequence so ordinary that it is easy to overlook: voters see a face before they read a platform. A visual impression arrives almost automatically, whereas learning a candidate's ideology and priorities requires time and effort. This ordering matters because judgments made from faces can form after exposures as brief as 100 milliseconds and predict real election outcomes (Todorov et al. 2005). The familiar contrast between "faces" and "facts" is therefore not merely rhetorical. It describes two sources of political judgment that become available at different speeds and costs.

Time pressure matters because it changes how decision makers acquire and combine information before choosing. When time is scarce, people accelerate processing, filter cues, or substitute a simpler noncompensatory rule for effortful integration (Payne, Bettman, and Luce 1996; Rieskamp and Hoffrage 2008). Political learning follows the same basic logic. Candidate evaluations can be updated on-line as information is encountered, even when the facts supporting the resulting impression are poorly remembered (Lodge, McGraw, and Stroh 1989). A deadline can therefore alter which evidence enters the running evaluation and how extensively competing considerations are compared before a vote.

Antonakis and Dalgas (2009) make this puzzle unusually concrete. After an imaginary voyage from Troy to Ithaca, Swiss children aged 5–13 were shown pairs of photographs from French parliamentary runoffs and asked whom they would choose as captain. Their choices predicted the actual winner with a probability of 0.71; adults' competence ratings predicted the winner with a probability of 0.72, and the

child and adult response patterns were statistically indistinguishable. The participants did not need campaign knowledge, policy positions, or electoral experience to reproduce the electorate's ranking. As Antonakis and Dalgas put it, "voters are anchored in first impressions and do not appropriately correct initial inferences" (2009, 1183). Ballew and Todorov (2007) identify the speed of this process more directly: competence judgments after 100 or 250 milliseconds were as predictive as judgments made with unlimited exposure, and a two-second response deadline preserved predictive accuracy while instructions to deliberate lengthened response time and reduced it.

That finding is compelling, but it leaves the central decision process unobserved. We know that faces can predict choices; we know much less about when voters stop at the face, when they seek political information, and whether constraints make one source dominate the other. Candidate photographs operate as cues in low-information elections (Banducci et al. 2008), appearance predicts success across political contexts (Rosar et al. 2008; Berggren et al. 2010; Lawson et al. 2010), and media exposure can strengthen appearance-based voting among less informed citizens (Lenz and Lawson 2011). Faces also convey perceived competence, ideology, party, status, and threat rather than attractiveness alone (Mattes et al. 2010; Herrmann and Shikano 2016; Olivola et al. 2018).

1.1 Gendered scrutiny and the cost of learning

Candidate gender can shape not only the final vote but the path by which voters learn. Ditonto, Hamilton, and Redlawsk (2014) show that voters seek more competence-related and "compassion issue" information about women than men. Andersen and Ditonto (2020) further show that the amount of search triggered by a woman candidate depends on partisan alignment and is strongest early in a campaign. This is consequential because women are often required to provide stronger evidence of qualification: across survey experiments, voters applied more stringent qualification standards to women candidates (Bauer 2020). A photograph makes candidate gender immediately available before policy text is opened, exactly the sequence in which an early cue can organize subsequent search.

The mechanism need not involve conscious hostility. Campaign communication can activate gender stereotypes that otherwise remain dormant (Bauer 2015), and rapid appearance-based inferences predict different outcomes by candidate gender: competence is most predictive in men-only races, whereas attractiveness predicts women-only races (Ditonto and Mattes 2018). Bahamonde and Sarpila (2024) likewise find a compounded penalty for working-class-looking women and weaker electoral returns to attractiveness for women than men. Recent work cautions against a universal penalty: eye-tracking evidence finds similar attention to stereotype-consistent and inconsistent information across candidate gender, suggesting that bias can arise from the standards applied to evidence rather than from attention alone (Jenke 2025), while think-aloud evidence shows both rapid gender heuristics and later gendered rationalizations (Rohrbach and Schönhagen 2025). The empirical question is therefore not simply whether women lose. It is whether profiles of women candidates invite different patterns of information acquisition, whether constraints suppress potentially corrective learning, and whether the same evidence is translated into choice differently for women and men.

Taken together, these findings motivate a design that follows the decision process from initial exposure to information acquisition and, ultimately, choice. Every respondent encounters the same paired-candidate task, while random assignment determines whether the choice occurs under a nine-second deadline, whether political information is costly to access, or whether neither constraint applies. Candidate-specific information requests reveal whether these constraints alter both the amount and the gendered distribution of political learning. The study thus locates inequality within a sequence—from first impression, to information acquisition, to choice—rather than only in the final vote. Methodologically, it combines randomized time and effort constraints with behavioral measures of information acquisition and repeated

choices between profiles constructed from Colombian electoral data.[1] Standard conjoint average marginal component effects require independently randomized attributes (Hainmueller, Hopkins, and Yamamoto 2014). Here, photographs, gender, ideology, electoral history, and facial characteristics remain bundled within real candidate profiles. The primary causal estimands therefore concern the randomized decision environment and the assignment of complete profiles. Analyses of individual profile characteristics assess heterogeneity in those contrasts, with the reported estimates further conditional on interview completion.

The first contribution is to recast appearance-based voting as a problem of information acquisition and evidence integration rather than simply an association between facial impressions and electoral choice. The same observed vote can emerge from several distinct processes: a voter may never consult political information, may inspect only one candidate's positions, may examine both candidates yet retain the initial visual ranking, or may use programmatic evidence to revise that ranking. These paths carry different theoretical and practical implications, but studies that observe only trait judgments or final votes cannot distinguish among them. By measuring candidate-specific information requests, decision time, and choice across repeated tasks, the experiment separates the decision to acquire political information from the electoral decision made in its presence or absence. The comparison among a deadline, an access cost, and an unconstrained baseline then reveals whether temporal and effort constraints reduce learning altogether, make learning asymmetric across candidates, or alter choice even when average acquisition changes little. The design therefore moves beyond showing that faces matter: it identifies the decision environments under which voters acquire evidence capable of supplementing an initial visual impression and estimates whether those environments alter candidate choice.

The second contribution is to connect this process account to gender inequality in candidate evaluation. More extensive information search about women candidates may reflect genuine openness to learning, but it may also indicate that women must supply additional evidence before voters regard them as qualified; conversely, limited search may signal either confidence or exclusion from serious consideration (Ditonto, Hamilton, and Redlawsk 2014; Bauer 2020). A gender difference in information acquisition therefore has no invariant meaning apart from the conditions under which it arises and the choices that follow it. Randomizing the decision environment supplies the relevant leverage. If the difference between women and men candidates expands, contracts, or reverses under time pressure or costly access, the experiment identifies how those constraints redistribute scrutiny across fixed profile groups, without treating candidate gender as independently randomized. Linking candidate-specific acquisition to candidate selection further documents whether unequal scrutiny is accompanied by unequal electoral returns. This contribution is consequential beyond the experimental interface: campaigns, ballot environments, and digital information platforms routinely vary in the time and effort they require from voters. Establishing whether such frictions place unequal informational burdens on women candidates connects a micro-level process of political learning to broader inequalities in evaluation and representation.

2 Research design

2.1 Sample and eligibility

The planned sample consists of 462 adults aged 18–65, balanced across men and women and allocated approximately equally across the three experimental conditions (154 per condition).[2] To minimize candidate recognition and prior knowledge of Colombian politics, respondents must not be Colombian nationals, reside in Colombia, or have previously lived in Colombia. Country of residence and nationality are selected from country lists, followed by a question on previous residence in Colombia. Respondents

[1] The study is registered in the AEA RCT Registry, AEARCTR-0017574, version 1.3, https://doi.org/10.1257/rct.17574-1.3. IRB Approval Number 086-2025. This paper describes the protocol following pretesting, including the revised nine-second deadline and fifteen analytical tasks. The registration's sample-size fields specify 444 participants and eight tasks, whereas its abstract specifies 462 participants. The current protocol specifies 462 participants and randomizes complete profiles rather than individual attributes.

[2] Participants are recruited through the survey company Borde Data (https://www.bordedata.cl).

who report a Colombian connection or decline to answer the residence-history question leave the study before the choice tasks. Eligibility screening and the baseline questionnaire precede exposure to the assigned-condition instructions. The questionnaire measures age, gender identity, occupation, education, electoral participation, political interest, frequency of political discussion, and ideological self-placement. Self-placement uses a 1–10 scale, anchored at left and right, with separate options for uncertainty, nonresponse, and identification with neither side.

An eligible completed interview requires reaching the final study screen, submitting all fifteen analytical choice tasks, recording a valid candidate selection in at least thirteen of them, and no detected prohibited browser navigation. The same rule applies in all three conditions and allows at most two analytical tasks without a recorded choice. Practice-task selections do not determine completion. After the initial instructions, respondents are instructed not to refresh the browser or use browser-history navigation. Detected reloads or history navigation during subsequent instructions, practice, analytical tasks, or their follow-up questions are recorded for audit and terminate participation, classifying the response as incomplete. Detection is limited to events observable by the application and does not establish the respondent's intent. Follow-up questions are required when applicable, while responses omitted by the questionnaire's conditional structure remain undefined rather than constituting incomplete data. Reaching the final screen without meeting these requirements is therefore distinguished from completing an interview. All substantive analyses use only interviews meeting these criteria, while recruitment accounting separately reports screened-out, navigation-excluded, and other incomplete attempts.

2.2 Stimuli and repeated choice

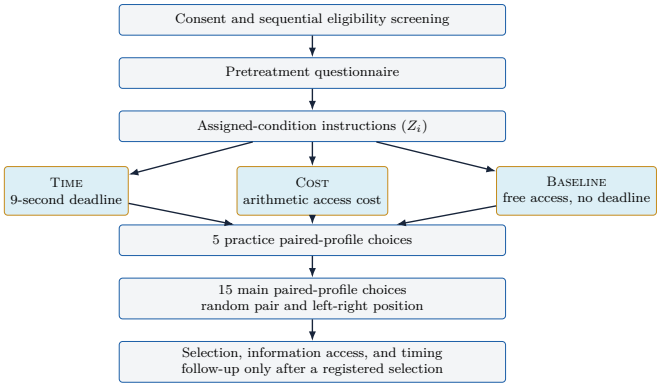
After the pretreatment questionnaire, respondents complete repeated paired choices drawn from 71 Colombian candidate profiles for which a photograph and political information are available. For each respondent, 40 profiles are sampled without replacement and randomly ordered. Consecutive profiles form 20 pairs, randomizing both pair composition and task order. The first five pairs are used for practice, and the remaining fifteen constitute the analytical sample. Within each task, the two candidates are assigned to left and right by an independent random permutation of the pair. No candidate therefore appears more than once for the same respondent, although candidates can appear for different respondents. Both photographs become visible together after loading. Descriptions are initially hidden and identify each candidate's ideological orientation as left, center, or right, followed by the candidate's thematic priorities. Respondents may reveal either description and revise their selection until they submit the task or the deadline expires. Information can therefore be accessed before or after an initial selection. The five practice rounds reproduce the assigned condition, including its deadline or arithmetic requirement, and are excluded from the analytical sample.

The fifteen-task analytical sequence is supported by evidence on the response quality of repeated conjoint choices. Across six online experiments involving 6,534 respondents recruited through Amazon's Mechanical Turk and Survey Sampling International, Bansak et al. (2018) examined thirty paired candidate choices per respondent. They found only limited increases in survey satisficing. The estimated effects of candidate education and partisanship declined slightly between the first and second tasks but remained broadly stable across the remaining sequence, as did the explanatory power of these attributes. Fifteen analytical choices, and twenty choices when practice is included, therefore fall within a range over which repeated candidate evaluations have retained substantial informational value. Because the appropriate task burden also depends on the full instrument, we will examine response times, information requests, nonresponse, and choice patterns across task positions to assess fatigue under the additional demands of photographic evaluation, experimental constraints, and follow-up questions.

After each analytical task with a registered selection, respondents are asked whether they would vote for the candidate they have just chosen in a real election. Those answering affirmatively identify the principal considerations underlying their choice. The eligible response options are randomized within each task, with "Otra razón" always placed last. The questions themselves retain a fixed order. The



A. Example choice task. Respondents compare 2 Colombian candidates and may open political information before submitting a selection. Information for the candidate on the left is shown here.



B. Assignment and measurement. Assignment is concealed until the condition-specific instructions and remains fixed during practice and analytical tasks. Each respondent receives an independent 40-profile sequence sampled without replacement from 71 eligible profiles. Pair order and left-right placement are randomized.

Figure 1: Experimental sequence and example choice task.

factor list includes appearance-related considerations, while ideology and thematic priorities appear only if at least one candidate description was revealed. Perceived time pressure is measured with the same seven-point item in Time and Baseline, regardless of the answer to the real-election support question. Perceived effort is measured on a seven-point scale in Cost only when the respondent revealed information. Tasks that end without a selection skip every follow-up question.

2.3 Experimental conditions

Let $Z_i \in \{T, C, B\}$ denote assignment to one of three decision environments. In Time, each task includes a visible nine-second countdown, while the candidate descriptions can be opened without an arithmetic requirement. In Cost, there is no deadline, but revealing each candidate's description requires correctly solving a simple addition problem. In Baseline, neither a deadline nor an arithmetic barrier applies. Allocation occurs with equal probability at the respondent level when study sessions are initialized and remains concealed through screening and the baseline questionnaire. The candidates, number of tasks, and choice prompt are otherwise held constant. This structure distinguishes two constraints on political learning. Time limits the time available to acquire and integrate information, whereas Cost increases the effort required to obtain it. In every condition, decision timing begins only when both photographs have loaded and decoded successfully and are revealed together. Loading time is recorded separately, and an image-loading failure prevents the task from starting until the images are available. The same nine-second deadline applies during practice and analytical tasks in Time.[3] If the deadline expires, the current selection is submitted automatically. When no candidate has been selected, the choice remains missing.

2.4 Temporal scarcity and deadline calibration

The time-pressure manipulation is calibrated to the demands of the decision task rather than to a universal threshold for rapid judgment. Ordóñez and Benson (1997, 122) distinguish the existence of a deadline from its psychological consequences, noting that "it is possible to have time constraint but no time pressure." A consequential deadline must therefore restrict the time respondents would otherwise use to acquire and evaluate information. Experimental evidence shows that temporal constraints can

[3]The assigned deadline is retained in browser storage when available. Participants are explicitly warned against refreshing or using browser history. Detected violations terminate participation and exclude the case from the analytical sample, while preserving previously recorded responses and the reason for termination. Stimulus readiness, deadline expiry, response preservation, and navigation handling were verified through isolated simulations and browser tests.

induce faster processing, more selective information search, and changes in decision strategy (Payne, Bettman, and Johnson 1988). Although Ballew and Todorov (2007) elicited electorally predictive competence judgments under a two-second deadline, their task involved photographs alone. The present task additionally permits respondents to reveal and compare political descriptions. Across the available candidate pairs, these descriptions contain a median of 28 words in total. Brysbaert's (2019) estimated Spanish silent-reading rate of 278 words per minute implies approximately six seconds of uninterrupted reading, before allowing for information access, comparison, and candidate selection. This calculation provides context rather than a minimum comprehension time. Moreover, cursor movement and target selection depend on interface geometry and input devices, making fixed allowances per click inappropriate (MacKenzie 1992). The relevant benchmark is consequently respondents' observed performance on the experimental task itself.

The nine-second deadline is informed jointly by research on task-relative time constraints and evidence from a pre-test of the experimental task. The pre-test evaluated an exploratory twenty-second deadline against an unconstrained baseline to assess how strongly that ceiling restricted task-completion time. The pre-test comprised 30 eligible completed interviews, contributing 450 observations across 15 analytical tasks per respondent. Panel A of Figure A1 reports mean task duration by task position among these completed pre-test participants. Across these tasks, pooled mean duration was 13.68 seconds (SD 14.64, 13 respondents and 195 task observations) in the unconstrained condition and 9.99 seconds (SD 4.28, 10 respondents and 150 task observations) under the exploratory 20-second deadline. The arithmetic-cost condition contributed the remaining 7 respondents and 105 task observations. The corresponding unconstrained median was 10.47 seconds (interquartile range 6.43–15.27), and only 15.9 percent of unconstrained tasks (31 of 195) lasted longer than 20 seconds. These observations suggest that the exploratory ceiling frequently exceeded the time respondents used without a deadline, although they do not establish an absence of experienced pressure. To impose a more binding restriction, we adapt the calibration principle employed by Kocher et al. (2019, 220–221), whose deadlines represented "a 20% reduction of the median decision times in the absence of constraints." Their restrictions applied to blocks of choices rather than individual tasks, so the present design adopts their relative calibration principle rather than their implementation. Applying that reduction to the unconstrained median yields 8.38 seconds, rounded upward to nine seconds, approximately 14 percent below the observed median. Rounding upward retains a modest margin for information access and response registration while keeping the deadline below typical unconstrained task duration. Although nine seconds exceeds the unconstrained median time to first selection of 8.39 seconds, it remains below the typical time used to complete the full task. This restriction therefore compresses opportunities for information acquisition and comparison without assuming that political information becomes inaccessible. Its effectiveness will be assessed jointly through reported hurry, information acquisition, and nonselection at the deadline, rather than inferred from mechanically shortened task durations alone. These completion-conditioned pre-test statistics inform calibration but do not establish that nine seconds is optimal.

2.5 Outcomes and measurement

The primary outcomes are recorded candidate selection, whether any description is revealed, the number of descriptions revealed, time to first selection capped at a common nine-second window, and failure to register a selection within that window. Description revelation is the observed measure of information acquisition and does not establish that the text was read or understood. These candidate-specific records permit tests of gendered scrutiny. Total time from photograph display to task submission, changes in selection, and information-button presses are recorded separately. Information-acquisition measures cover the entire task, including openings after an initial selection. Terminal nonselection at the deadline is retained as a separate outcome. Post-choice measures capture whether respondents would support the selected candidate in a real election and which considerations they report using. The candidate profiles include gender, age, party affiliation, geographic constituency, electoral performance, political

information, and previously measured facial characteristics. Recording the alternatives and information available in each task allows us to distinguish treatment effects on information acquisition from treatment effects on recorded selection.

3 Potential outcomes and estimands

3.1 Notation and common averaging distribution

For eligible respondent i in analytical task t, let $Z_i \in \{T, C, B\}$ denote the assigned decision environment and let $P_{it} = (C_{it}^L, C_{it}^R)$ denote the ordered candidate pair. For side $s \in \{L, R\}$, define $M_{it}^s(z, p) \in \{0, 1\}$ as the potential indicator that the respondent reveals the political description of the candidate on side s at any point before the task ends under environment z and pair p. This measures access to the text, without establishing that it was read or understood. Respondents may reveal information after an initial selection and revise that selection before submission. Aggregate information acquisition is defined as

$$K_{it}(z, p) = M_{it}^L(z, p) + M_{it}^R(z, p), \quad A_{it}(z, p) = \mathbf{1}\{K_{it}(z, p) \geq 1\}, \quad X_{it}(z, p) = \mathbf{1}\{K_{it}(z, p) = 2\}.$$

Thus, K is the number of descriptions revealed, A indicates access to at least one description, and X indicates access to both descriptions.

Let $T_{it}^*(z, p)$ denote seconds from the simultaneous display of both fully loaded photographs to the first candidate selection, with $T^* = \infty$ if no selection occurs. To place all conditions on a common observation window, define capped selection latency $D_{it}^9(z, p) = \min\{T_{it}^*(z, p), 9\}$ and slow or absent selection $S_{it}^9(z, p) = \mathbf{1}\{T_{it}^*(z, p) > 9\}$. If no selection occurs within nine seconds, $D^9 = 9$ and $S^9 = 1$. This common analytical window does not impose a deadline on the baseline or arithmetic-cost conditions, whose uncapped response times remain recorded. The full task duration, measured from photograph display to submission, is recorded separately and can include information acquisition and revisions after the first selection. Loading time precedes both measures. Let $Q_{it}(z, p)$ indicate that a task terminates at the deadline without any candidate selection. This outcome is possible under TIME; the other conditions require a selection before submission. Reaching the deadline after selecting a candidate preserves the current selection and gives $Q = 0$. Thus, S^9 measures failure to select within the common window, while Q identifies termination without a candidate response.

Candidate choice is defined without privileging the randomized left position. Let $V_{it}(z, p)$ denote the candidate selected when the task ends; $V = \varnothing$ denotes a missing candidate response when $Q = 1$. For each displayed side, define the registered-selection event

$$Y_{it}^s(z, p) = \mathbf{1}\{V_{it}(z, p) = C_{it}^s\}.$$

When a candidate response is recorded, $Y_{it}^L + Y_{it}^R = 1$; when $Q_{it} = 1$, both derived event indicators equal zero, while the candidate response itself remains missing. This definition neither imputes a vote nor assigns an unobserved preference to a respondent who made no selection. In the full eligible randomized population, its expectation is the probability that a displayed profile receives a registered selection under the assigned environment. Effects on this outcome can reflect both changes in which candidate is selected and changes in whether any selection is completed.

Unless otherwise stated, the causal estimands below weight eligible respondents and the fifteen analytical tasks equally and average over the randomized candidate draw, pairing, task order, and left-right assignment. Practice tasks are excluded. Comparisons between $z \in \{T, C\}$ and BASELINE define respondent-level intention-to-treat targets. The reported estimates are restricted to completed interviews and must be distinguished from these full-population targets, as detailed in the estimation strategy.

3.2 Effects of the decision environment

For each task-level outcome $R \in \{A, K, X, D^9, S^9\}$, define

$$\tau_{zB}^R = \mathbb{E}[R_{it}(z, P_{it}) - R_{it}(B, P_{it})], \qquad z \in \{T, C\}.$$

The terminal-timeout effect is defined separately as $\tau^{Q}_{TB} = \mathbb{E}[Q_{it}(T, P_{it}) - Q_{it}(B, P_{it})]$ because forced task termination exists only in TIME. These estimands capture how temporal scarcity and information cost change information acquisition, decision speed, and response completion. Information acquisition is also a potential mediator of candidate choice. Because it occurs after treatment, models that condition choice on realized information acquisition do not identify a controlled or natural direct effect without additional assumptions; mediation analyses are therefore interpreted cautiously and do not carry a direct-effect interpretation.

Randomized pairing and position also identify side-neutral profile-selection probabilities. For complete candidate profile c, define

$$\theta_c(z) = \mathbb{E}[Y^s_{it}(z, P_{it}) \mid C^s_{it} = c], \qquad \Delta_c(z) = \theta_c(z) - \theta_c(B).$$

The expectation averages over respondents, tasks, randomized rivals, and side. $\Delta_c(z)$ is the causal effect of the decision environment on the probability that profile c is selected when displayed. Contrasts $\theta_c(z) - \theta_{c'}(z)$ compare randomly assigned complete profiles under the same environment. They do not isolate gender, ideology, party, or facial characteristics because those attributes remain bundled within each observed candidate.

3.3 Gendered information acquisition and choice

Let $G(c) = 1$ denote a profile coded as a woman. For $R \in \{M, Y\}$, define the candidate-level mean, gender-indexed contrast, and treatment-induced change as

$$\mu^R_g(z) = \mathbb{E}[R^s_{it}(z, P_{it}) \mid G(C^s_{it}) = g], \quad \Gamma_R(z) = \mu^R_1(z) - \mu^R_0(z), \quad \Psi_R(z) = \Gamma_R(z) - \Gamma_R(B).$$

$\Gamma_M(z)$ compares the probability of consulting political information about women and men candidates, while $\Gamma_Y(z)$ compares their probability of selection. The corresponding $\Psi_M(z)$ and $\Psi_Y(z)$ estimands identify how random assignment to a constrained decision environment changes each gender-indexed contrast. Candidate gender is not independently randomized, so $\Gamma_M(z)$ and $\Gamma_Y(z)$ compare complete profiles grouped by gender and are not causal effects of gender. The causal interpretation of $\Psi_M(z)$ and $\Psi_Y(z)$ is limited to the effect of the randomized environment on those fixed profile-group differences.

3.4 Hypotheses and analytical priorities

The **primary process family** consists of τ^A_{zB}, τ^K_{zB}, $\tau^{D^9}_{zB}$, and $\tau^{S^9}_{zB}$ for $z \in \{T, C\}$. The **primary gender-interaction family** consists of $\Psi_M(T)$, $\Psi_M(C)$, $\Psi_Y(T)$, and $\Psi_Y(C)$. The baseline contrasts $\Gamma_M(B)$ and $\Gamma_Y(B)$ form a separate descriptive family because candidate gender is not independently assigned. Hypothesis tests are two-sided even where theory implies a direction. Their empirical counterparts are estimated among completed interviews, so the causal status of the target estimands does not by itself establish a causal interpretation for the reported contrasts.

We expect both constrained conditions to reduce any information acquisition and the number of descriptions consulted, $\tau^A_{zB} < 0$ and $\tau^K_{zB} < 0$ (H1). We also expect TIME to reduce capped selection latency, $\tau^{D^9}_{TB} < 0$, while effects on S^9 and the timing effects of COST are tested without directional restrictions (H2). Under BASELINE, we expect respondents to consult information more often for profiles of women candidates, $\Gamma_M(B) > 0$, and both constraints to attenuate this difference, $\Psi_M(T) < 0$ and $\Psi_M(C) < 0$ (H3). Finally, we expect $\Psi_Y(T)$ and $\Psi_Y(C)$ to differ from zero, with their signs left open because prior work implies both potential penalties and advantages for women candidates (H4).

Additional estimands include acquisition of both descriptions X, terminal timeout Q, full task duration, profile-specific $\theta_c(z)$ and $\Delta_c(z)$, the real-election support item, self-reported time pressure and information cost, ideological congruence, associations between choice and other candidate characteristics, and moderation by respondent gender or political sophistication. **Additional analyses** examine mediation through information acquisition, models conditioned on realized information acquisition or completed choice, high-dimensional profile interactions, decision-factor responses, and subgroup variation.

Because these analyses require further assumptions or extend beyond the central hypotheses, they will be interpreted separately from the primary tests.

3.5 Timeout and missing-data rules

For respondents meeting the interview-completion criteria, all descriptions revealed before a task terminates contribute to M^s, A, K, and X, including when no candidate is selected. A task with no selection within nine seconds receives $D^9 = 9$ and $S^9 = 1$. If the TIME task terminates without a selection, $Q = 1$, the recorded candidate response remains missing, the derived registered-selection indicators satisfy $Y^L = Y^R = 0$, and every follow-up question is skipped. These undefined follow-up responses will not be imputed. If a selection has already occurred when the deadline expires, the current candidate response is retained, $Q = 0$, and the usual follow-up questions are shown. Up to two missing analytical choices are permitted within a completed interview, and their associated information and timing outcomes remain in the analysis. Participation ends when a third analytical task is submitted without a valid selection, because the minimum response requirement can no longer be met. These respondents, and those who otherwise do not finish the study, are excluded from substantive analyses but remain in recruitment accounting, with their previously recorded responses preserved. Withdrawal or abandonment before task submission is not recoded as a timeout. No missing candidate choice or undefined follow-up response will be imputed.

4 Estimation strategy

All substantive analyses are restricted to eligible completed interviews as defined above. Each retained respondent contributes all fifteen analytical tasks, including up to two tasks without a registered candidate selection. Outcomes A, K, D^9, and S^9 are estimated using linear models with indicators for TIME and COST, giving unadjusted contrasts relative to BASELINE within this sample. Let $H_i = 1$ indicate that respondent i meets the observed completion criteria. For a task-level outcome R, the reported condition contrast is

$$\delta^R_{zB} = \mathbb{E}[R_{it} \mid Z_i = z, H_i = 1] - \mathbb{E}[R_{it} \mid Z_i = B, H_i = 1].$$

Because the decision environment may affect reaching the final screen, recording at least thirteen choices, or detected browser navigation, conditioning on $H_i = 1$ can select different respondents across conditions. Consequently, δ^R_{zB} need not equal the full-population intention-to-treat effect τ^R_{zB}, even under randomized assignment. The reported contrasts are therefore interpreted as completion-conditioned comparisons rather than automatically as causal effects for all eligible randomized respondents. Screening, final-screen arrival, navigation-related exclusions, and other incomplete interviews are reported separately to make this selection transparent. Standard errors are clustered by respondent. Covariate-adjusted specifications using pretreatment characteristics and task-order indicators may improve precision but do not, by themselves, remove completion-related selection.

Candidate-level analyses use two observations per analytical task among completed interviews, one for each displayed candidate, including retained tasks that terminate without a selection. Linear probability models for M^s and the derived registered-selection outcome Y^s include treatment indicators, candidate gender, treatment-by-gender interactions, randomized side, and task-order indicators. Their interaction coefficients are completion-conditioned empirical counterparts of $\Psi_M(z)$ and $\Psi_Y(z)$ and share the same selection limitation. Primary uncertainty for these models uses two-way clustering by respondent and candidate. Profile-specific selection probabilities likewise condition on completed interviews and are not interpreted as attribute-level average marginal component effects.

Supplementary analyses describe which candidate characteristics are associated with registered selection in each decision environment. For gender, displayed ideology, dominance classification, and candidate age category, we report both the proportion selected among displayed profiles and the composition of the selected profiles. The former accounts for how often each characteristic is encountered, whereas the latter describes the candidates respondents choose. A joint linear probability model includes

these four classifications, their interactions with the assigned condition, randomized side, and task-order indicators. Within-condition contrasts compare women with men, left- and right-identified profiles with centrist profiles, high with low dominance, and older with younger age categories. These are adjusted associations across observed profile bundles rather than causal effects of independently assigned attributes. The model does not condition on whether respondents opened political information, which is itself a response to the experimental environment. Two-way clustered confidence intervals and unadjusted two-sided tests accompany these exploratory contrastsin Appendix B, separately from the primary hypothesis families.

Covariate-adjusted predictions complement these unadjusted comparisons without replacing the primary tests. Separate logistic working-mean models describe any information acquisition, acquisition of both descriptions, failure to select within the common observation window, candidate-specific information acquisition, and registered candidate selection. Mean-bias-reduced estimation yields finite logistic estimates in the presence of separation, which can arise when every member of a small subgroup opens information (Kosmidis and Firth 2021). A fractional-logit model with a quasi-binomial variance describes the proportion of descriptions acquired, with predictions rescaled to the expected number of descriptions. Capped selection latency is modeled by ordinary least squares and reported in seconds rather than as a probability. Every specification adjusts for respondent age, gender, completion of tertiary education, baseline political interest, and task position. Candidate-level models additionally include randomized side and treatment-by-candidate-gender interactions. An exploratory information-acquisition model adds interactions with respondent gender. No model controls for realized information acquisition, response time, reported hurry, or arithmetic success when these variables are consequences of the assigned environment.

For each decision environment, fitted outcomes are averaged over the same observed covariate distribution rather than evaluated for an artificial average respondent (Arel-Bundock, Greifer, and Heiss 2024). Thus, for a task-level outcome R, the adjusted mean is

$$\widehat{\mu}^{R}_{\text{adj}}(z) = \frac{1}{nT} \sum_{i:H_i=1} \sum_{t=1}^{T} \widehat{m}_R(z, X_i, t),$$

where n is the number of eligible completed respondents, T is the number of analytical tasks, and X_i contains pretreatment characteristics. Candidate-level predictions change only the decision environment and retain the observed profile bundles, averaging separately within candidate-gender groups. Respondent-gender panels likewise retain the observed members of each subgroup. Approximate pointwise confidence intervals use respondent-clustered covariance for task-level models and two-way respondent–candidate clustering for candidate-level models. Probability intervals are calculated on the logit of the average predicted probability and transformed back, preserving the empirical averaging target and the probability bounds. Student-t reference distributions are used, with the smaller cluster count determining degrees of freedom in two-way analyses. These model-based comparisons remain conditional on completion and on the chosen working specification. Neither adjustment nor nonlinear estimation removes completion-related selection or identifies an isolated causal effect of candidate gender.

Within the primary process and gender-interaction families, two-sided p-values are adjusted separately by Holm's sequential procedure. The two baseline gender contrasts constitute a third, descriptive multiplicity family. Unadjusted estimates with 95% confidence intervals are also reported. Pretreatment characteristics and the distribution of candidate profiles and positions are summarized within the completed sample. Manipulation checks examine arithmetic-problem completion, information acquisition, D^9, S^9, and Q, together with reported hurry in Time relative to Baseline among tasks with a registered selection. These checks are conditional on interview completion, and the hurry comparison additionally conditions on an observed answer. The timeout frequency among completed interviews excludes respondents with more than two missing choices and therefore does not represent the timeout frequency of all recruits. Task-order analyses also use only retained interviews, while practice tasks

remain outside the analytical sample. Candidate gender and respondent gender are distinguished throughout.

5 Preliminary tables and figures

Incomplete fieldwork. Tables report descriptive statistics, regression estimates, pointwise 95% confidence intervals, and two-sided tests. Except for recruitment accounting, every table and figure uses only eligible completed interviews. Daily updates are interim monitoring summaries and do not define a significance-based stopping rule.

Table 1: Recruitment and analytical coverage

Stage	N
Survey entrants across study versions	110
Screened out	3
Initial screening or baseline not completed	33
Eligible participants not yet finished	11
Finished below the response threshold	0
Excluded after detected reload or browser-history navigation	4
Stopped after exceeding the permitted missing choices	1
Other terminated incomplete cases	0
Other-protocol completes excluded from analysis	33
Eligible completes included in analysis	25

Notes. An eligible completed interview reaches the final screen, submits all 15 analytical tasks, and records at least 13 valid choices, with no detected prohibited browser navigation. Detected navigation exclusions are reported separately. Recruitment accounting covers all started attempts across study versions. The cumulative total comprises 58 eligible completes, of which 33 belong to other protocol versions and are excluded from the current analyses. Screened-out respondents and incomplete attempts do not enter substantive analyses. The analytical sample contains 25 eligible completed interviews.

Table 2: Completed interviews and analytical choices by assigned condition

Condition	Respondents	Tasks	Valid choices	Missing choices
Baseline	9	135	135	0
Time	8	120	112	8
Cost	8	120	120	0

Notes. The 25 eligible completed interviews contribute 375 analytical tasks. Each interview contributes 15 submitted analytical tasks and at least 13 valid candidate choices. Missing choices remain missing in the original response and are retained in the information and timing denominators. Practice tasks are excluded.

Table 3: Pretreatment characteristics of eligible completed interviews

Characteristic	Category	Overall	Baseline	Time	Cost	Missing
n		25	9	8	8	
Age		37.5 (12.6)	41.4 (15.0)	34.0 (7.9)	36.5 (13.7)	0.0
Gender (%)	Woman	9 (36.0)	3 (33.3)	2 (25.0)	4 (50.0)	0.0
	Man	16 (64.0)	6 (66.7)	6 (75.0)	4 (50.0)	
	Another identity	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	
	No response	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	
Education (%)	Basic or less	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0.0
	Secondary	3 (12.0)	2 (22.2)	0 (0.0)	1 (12.5)	
	Technical or university	16 (64.0)	5 (55.6)	6 (75.0)	5 (62.5)	
	Postgraduate	6 (24.0)	2 (22.2)	2 (25.0)	2 (25.0)	
Political interest (1-7)		5.0 (1.9)	4.7 (1.7)	5.2 (2.4)	5.0 (1.9)	0.0
Left-right position (1-10)		3.8 (2.7)	5.1 (2.8)	2.9 (2.3)	3.1 (2.6)	8.0

Notes. The sample comprises 25 eligible completed interviews. Continuous entries are means with standard deviations in parentheses. Categorical entries are counts and percentages. Each respondent is counted once and meets the interview-completion criteria. Missing numeric self-placement includes uncertainty, nonresponse, and identification with neither side. The final column reports the overall percentage missing.

Table 4: Task-level descriptive statistics by decision environment

Measure	Baseline	Time	Cost
Respondents	9	8	8
Tasks	135	120	120
Any information (%)	100.00	82.50	87.50
Descriptions (mean)	1.95	1.61	1.67
Both descriptions (%)	94.81	78.33	79.17
Capped first-selection time (s)	7.26	4.87	8.28
No selection by 9 s (%)	57.04	6.67	82.50
Terminal nonselection (%)	0.00	6.67	0.00
Expired deadlines	0	19	0
Task duration (median s)	11.03	5.90	28.24

Notes. The 375 analytical tasks come only from 25 eligible completed interviews and exclude practice. Information acquisition covers the entire choice task. Capped time concerns the first selection, not submission, and uses a 9-second window. The permitted terminal nonselections remain in information and timing denominators. An expired timer can retain a choice and is not necessarily a nonselection.

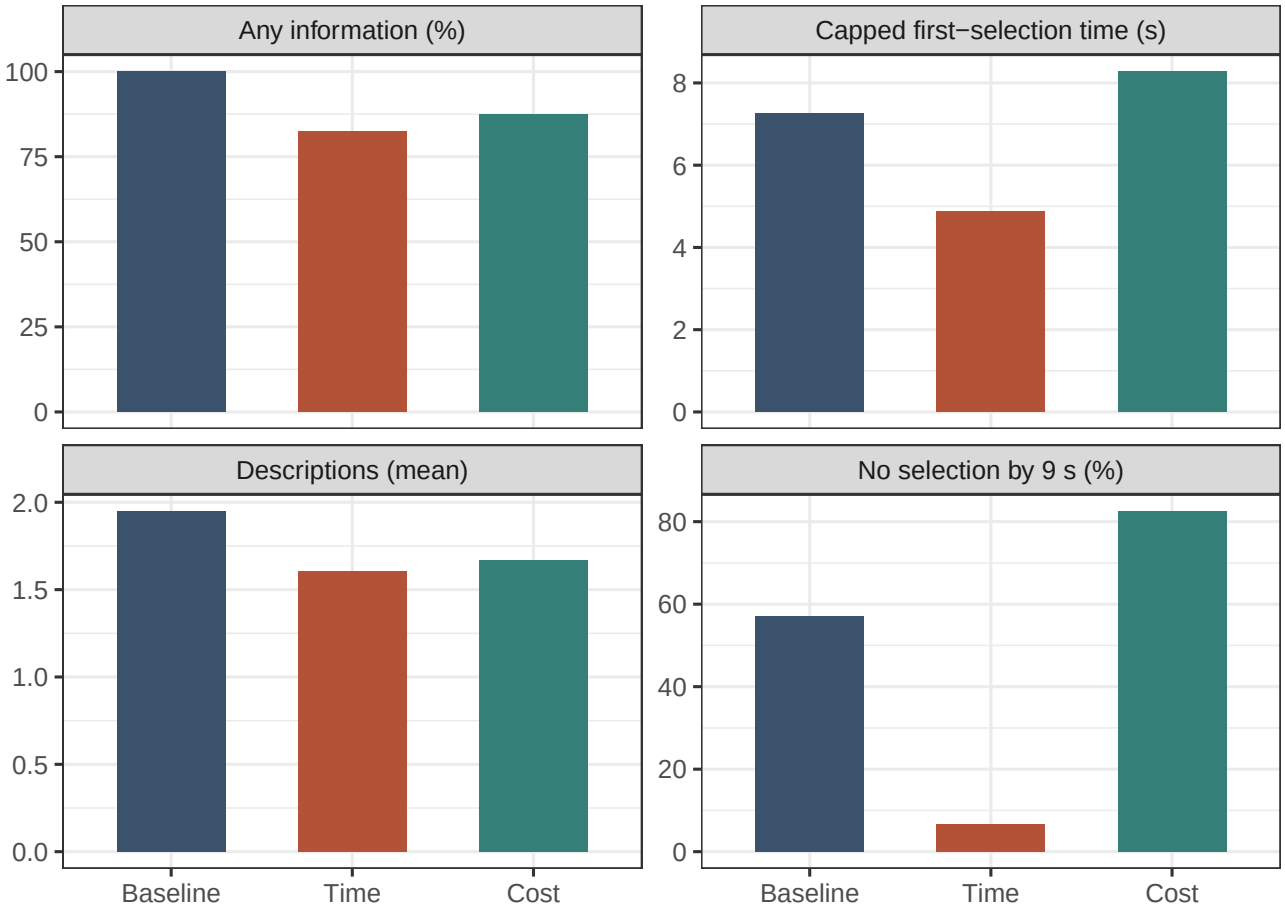


Figure 2: Observed condition means for the primary process outcomes. Means use 375 analytical tasks from 25 eligible completed interviews. Panels retain their natural units and use different vertical scales.

Table 5: Timing, perceived pressure, and information-cost checks

Measure	Baseline	Time	Cost
First selection (median s)	9.95	4.64	25.91
Full task duration (mean s)	16.96	5.93	34.23
Full task duration (median s)	11.03	5.90	28.24
Full task duration (75th percentile s)	19.38	7.53	39.54
Full task duration (90th percentile s)	36.06	9.00	55.06
Full task duration (95th percentile s)	51.83	9.00	91.37
Hurry responses (n)	135	112	0
Reported hurry (1-7)	1.96	4.31	–
Cost responses (n)	0	0	105
Reported effort (1-7)	–	–	3.77
Arithmetic attempts (mean)	–	–	1.91
Arithmetic passed (%)	–	–	87.50

Contrast	Estimate	95% CI	p
Hurry: Time - Baseline	2.35	[0.832, 3.867]	0.0047

Notes. The sample includes 25 eligible completed interviews and 375 analytical tasks. Timing measures describe the choice page, not the duration of the full study or its follow-up questions. Hurry is asked in the deadline and baseline conditions after a registered choice. Effort is asked in the cost condition after information is revealed. Follow-up means condition on an observed answer. Arithmetic success refers to at least one successful access during a cost-condition task. The hurry contrast uses 247 observed answers from 17 respondents, respondent-clustered 95% confidence intervals, and an unadjusted two-sided test. Undefined items remain missing. Untimed baseline durations provide an unrestricted descriptive reference for the task. Deadline-condition durations are mechanically bounded and cannot independently calibrate an appropriate deadline. Shorter durations in that condition alone do not establish experienced pressure.

Table 6: Primary process contrasts among completed interviews

Outcome	Contrast	Estimate	95% CI	p	Holm p
Any information	Time - Baseline	-0.175	[-0.371, 0.021]	0.0776	0.4656
Any information	Cost - Baseline	-0.125	[-0.372, 0.122]	0.3066	0.6773
Descriptions revealed	Time - Baseline	-0.340	[-0.785, 0.105]	0.1283	0.6414
Descriptions revealed	Cost - Baseline	-0.281	[-0.765, 0.202]	0.2414	0.6773
Capped first-selection time (s)	Time - Baseline	-2.390	[-4.013, -0.767]	0.0057	0.0396
Capped first-selection time (s)	Cost - Baseline	1.015	[-0.670, 2.699]	0.2258	0.6773
No selection within 9 seconds	Time - Baseline	-0.504	[-0.786, -0.221]	0.0012	0.0094
No selection within 9 seconds	Cost - Baseline	0.255	[-0.105, 0.614]	0.1568	0.6414

Notes. Unadjusted linear models use 375 analytical tasks from 25 eligible completed interviews, with 15 tasks per interview. Standard errors are clustered by respondent. Intervals are pointwise 95% intervals. The two-sided p-values are shown unadjusted and Holm-adjusted across the 8 primary process contrasts. Binary-outcome coefficients are probability differences, description coefficients are counts, and timing coefficients are seconds. Every retained respondent contributes equally many tasks. Completion can be affected by treatment, so these completion-conditioned contrasts do not automatically identify the full intention-to-treat effects.

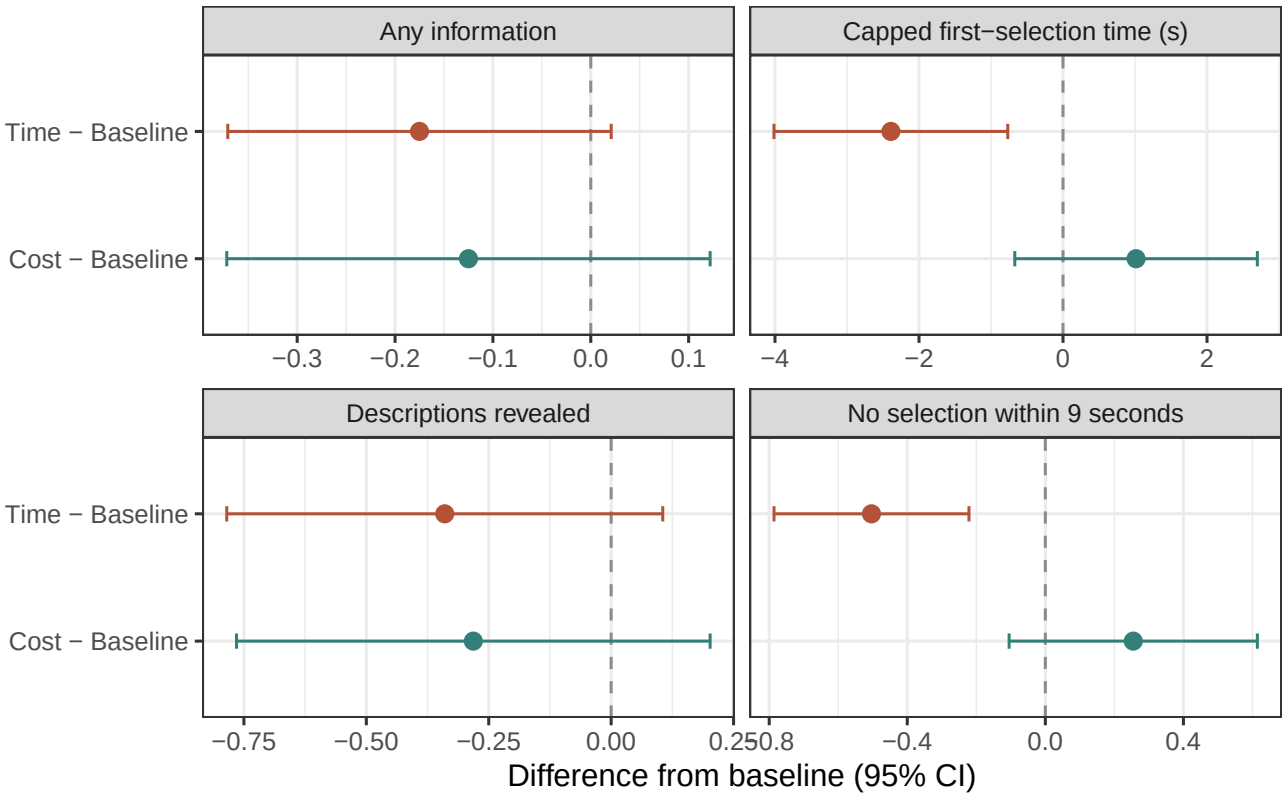


Figure 3: Direction and magnitude of the primary process contrasts. Estimates use 375 analytical tasks from 25 eligible completed interviews. The vertical line marks no difference from the unconstrained baseline. Bars show respondent-clustered pointwise 95% confidence intervals.

Table 7: Between-condition differences in gender-indexed profile contrasts

Outcome	Contrast	Estimate	95% CI	p	Holm p
Information acquisition	Time - Baseline	-0.119	[-0.234, -0.004]	0.042	0.17
Information acquisition	Cost - Baseline	-0.052	[-0.124, 0.021]	0.154	0.46
Registered selection	Time - Baseline	-0.003	[-0.208, 0.202]	0.975	1.00
Registered selection	Cost - Baseline	0.010	[-0.231, 0.251]	0.934	1.00

Notes. Completion-conditioned counterparts of $\Psi_M(z)$ and $\Psi_Y(z)$ come from models using 750 candidate appearances from 25 eligible completed interviews with condition, candidate gender, their interactions, randomized side, and task position. Standard errors are clustered by respondent and candidate. Intervals are pointwise 95% intervals. The two-sided tests are shown unadjusted and Holm-adjusted across the 4 primary interaction contrasts. The covariance calculation uses a positive-semidefinite correction when necessary. A submitted terminal nonselection contributes zero registered-selection events for both candidates, while the original vote remains missing. Gender is bundled with other candidate attributes. These are not gender AMCEs, and completion-related selection limits their causal interpretation.

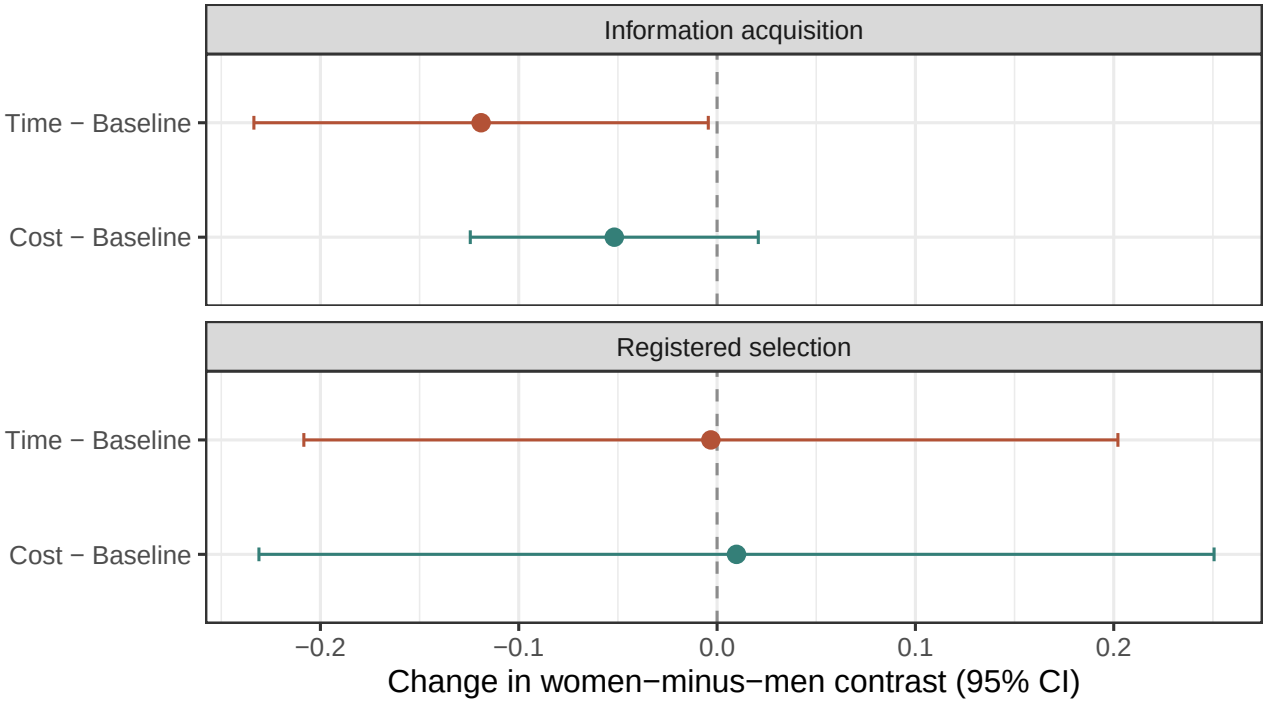


Figure 4: Direction of the gender-indexed condition interactions. Estimates use 750 candidate appearances from 25 eligible completed interviews. Positive values indicate a larger women-minus-men profile contrast than under baseline. Bars show two-way clustered pointwise 95% confidence intervals.

Table 8: Gender-indexed contrasts within each decision environment

Outcome	Condition	Gamma	95% CI	p	Holm p
Information acquisition	Baseline	0.006	[-0.022, 0.033]	0.676	1
Information acquisition	Time	-0.113	[-0.226, -0.000]	0.049	–
Information acquisition	Cost	-0.046	[-0.115, 0.023]	0.180	–
Registered selection	Baseline	0.040	[-0.088, 0.168]	0.522	1
Registered selection	Time	0.037	[-0.132, 0.206]	0.654	–
Registered selection	Cost	0.050	[-0.157, 0.257]	0.622	–

Notes. Entries give completion-conditioned counterparts of $\Gamma_M(z)$ and $\Gamma_Y(z)$ from the same candidate-level models using 750 candidate appearances from 25 eligible completed interviews, with two-way clustered pointwise 95% intervals. They compare complete profiles grouped by gender, not an isolated causal effect of gender. The 2 baseline descriptive tests form a separate Holm family. Other within-arm p-values are unadjusted descriptive comparisons.

Table 9: Additional pretreatment sample characteristics

Characteristic	Category	Overall	Baseline	Time	Cost	Missing
n		25	9	8	8	
Occupation (%)	Full-time paid work	10 (40.0)	3 (33.3)	4 (50.0)	3 (37.5)	0.0
	Part-time paid work	1 (4.0)	0 (0.0)	1 (12.5)	0 (0.0)	
	Self-employed	7 (28.0)	3 (33.3)	2 (25.0)	2 (25.0)	
	Student	2 (8.0)	1 (11.1)	0 (0.0)	1 (12.5)	
	Unpaid domestic or care work	1 (4.0)	1 (11.1)	0 (0.0)	0 (0.0)	
	Retired	1 (4.0)	1 (11.1)	0 (0.0)	0 (0.0)	
	Unemployed, seeking work	1 (4.0)	0 (0.0)	1 (12.5)	0 (0.0)	
	Unemployed, not seeking work	2 (8.0)	0 (0.0)	0 (0.0)	2 (25.0)	
	Other	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	
Voted last election (%)	Yes	21 (84.0)	8 (88.9)	7 (87.5)	6 (75.0)	0.0
	No	3 (12.0)	1 (11.1)	1 (12.5)	1 (12.5)	
	Do not recall	1 (4.0)	0 (0.0)	0 (0.0)	1 (12.5)	
Political discussion (%)	Never	2 (8.0)	1 (11.1)	0 (0.0)	1 (12.5)	0.0
	Less than weekly	5 (20.0)	3 (33.3)	2 (25.0)	0 (0.0)	
	Once or twice weekly	3 (12.0)	0 (0.0)	0 (0.0)	3 (37.5)	
	Several times weekly	7 (28.0)	2 (22.2)	2 (25.0)	3 (37.5)	
	Daily	8 (32.0)	3 (33.3)	4 (50.0)	1 (12.5)	
Residence (ISO) (%)	CL	22 (88.0)	7 (77.8)	7 (87.5)	8 (100.0)	0.0
	CR	1 (4.0)	0 (0.0)	1 (12.5)	0 (0.0)	
	PE	2 (8.0)	2 (22.2)	0 (0.0)	0 (0.0)	
Nationality (ISO) (%)	CL	21 (84.0)	7 (77.8)	7 (87.5)	7 (87.5)	0.0
	CR	1 (4.0)	0 (0.0)	1 (12.5)	0 (0.0)	
	PE	2 (8.0)	2 (22.2)	0 (0.0)	0 (0.0)	
	UY	1 (4.0)	0 (0.0)	0 (0.0)	1 (12.5)	

Notes. Counts and percentages refer to the 25 eligible completed interviews. ISO country codes identify residence and nationality.

5.1 Covariate-adjusted predictions

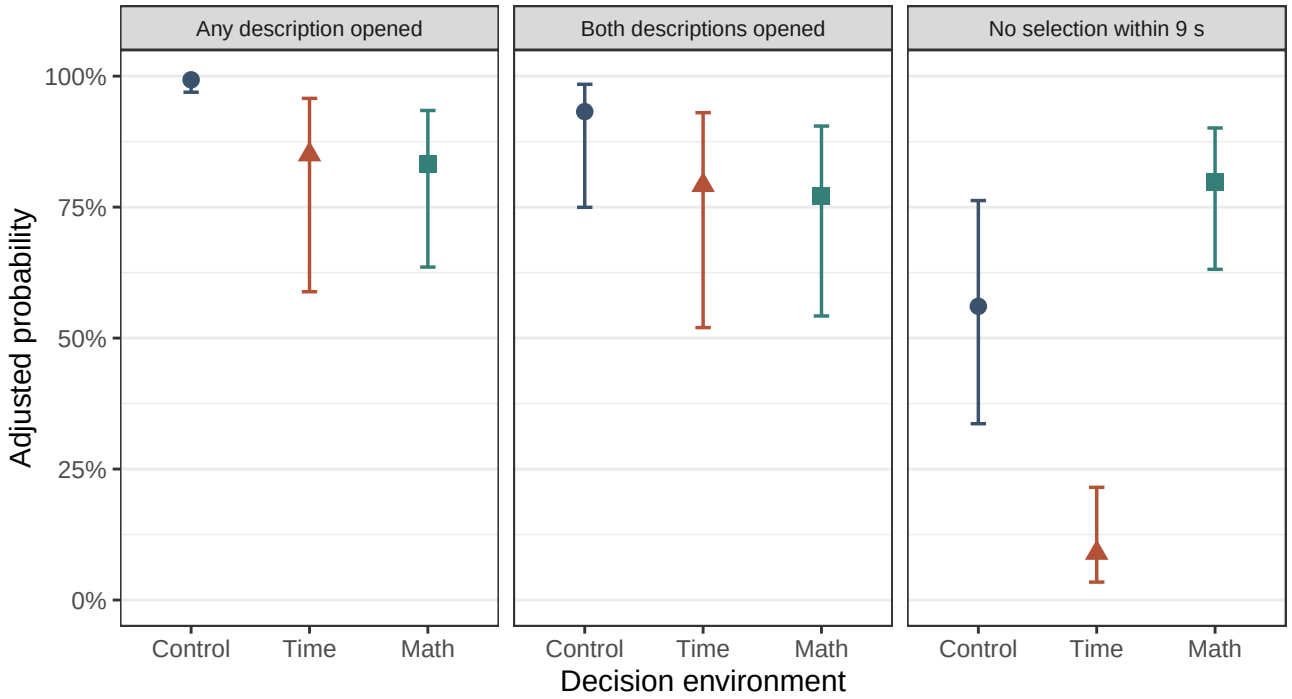


Figure 5: Adjusted probabilities of information acquisition and slow selection

Notes. Points report average predicted probabilities from separate mean-bias-reduced logistic regressions using 375 analytical tasks from 25 eligible completed respondents. All specifications adjust for age, respondent gender, tertiary education, baseline political interest, and linear task position. Each condition is evaluated over the same observed respondents and task positions. Bars are approximate pointwise 95% confidence intervals based on respondent-clustered covariance and 24 degrees of freedom. Intervals are calculated on the logit of the average probability and transformed back. Slow selection denotes no first selection within 9 seconds in any condition, not necessarily a terminal timeout. Permitted no-choice submissions remain in the sample. These are completion-conditioned model-based predictions, not unconditional intention-to-treat estimates.

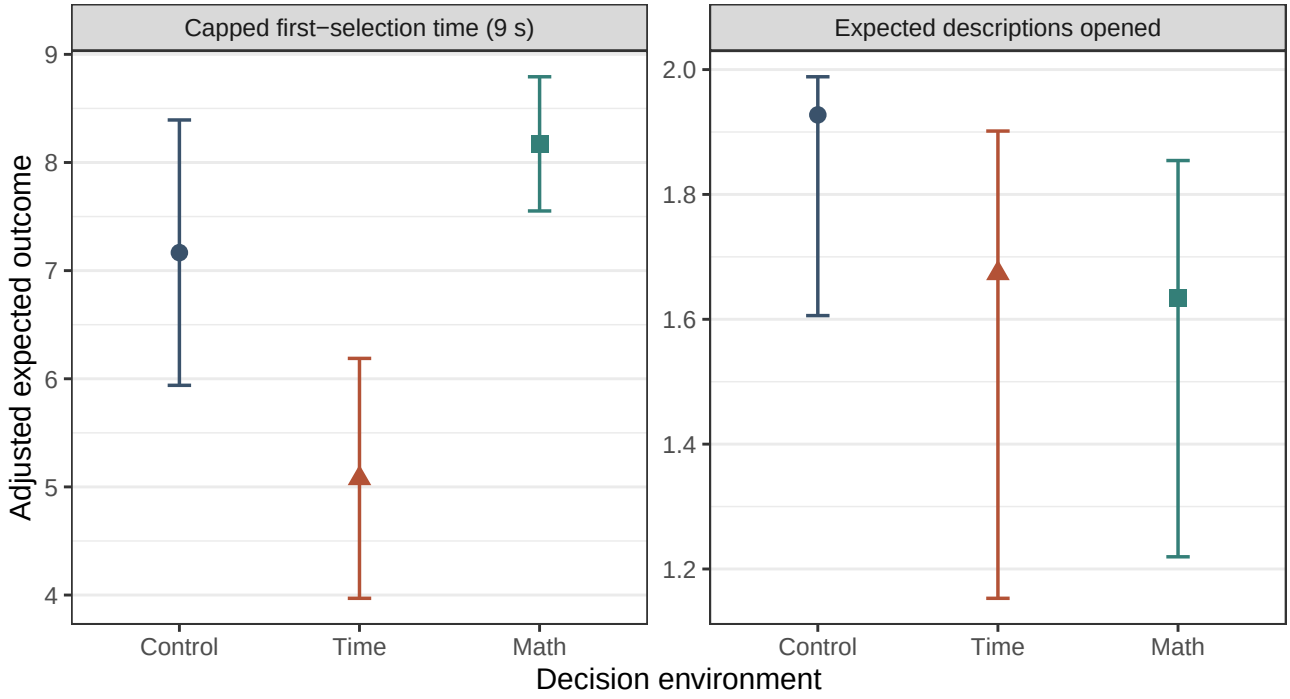


Figure 6: Adjusted expected information acquisition and first-selection latency

Notes. Expected descriptions are estimated by a fractional-logit quasi-binomial model for the share of the 2 available descriptions opened, then rescaled to a count. Capped first-selection latency is estimated by ordinary least squares in seconds, using the common 9-second window. The panels retain their different natural units and vertical scales. Models use 375 analytical tasks from 25 eligible completes and the pretreatment controls and empirical standardization described in Figure 5. Bars show approximate respondent-clustered pointwise 95% confidence intervals. The description-count intervals use a transformed probability scale before rescaling. A task with no selection within the common window contributes 9 seconds to capped latency. The figure concerns the first selection, not full task duration, which is reported separately in Figure A1.

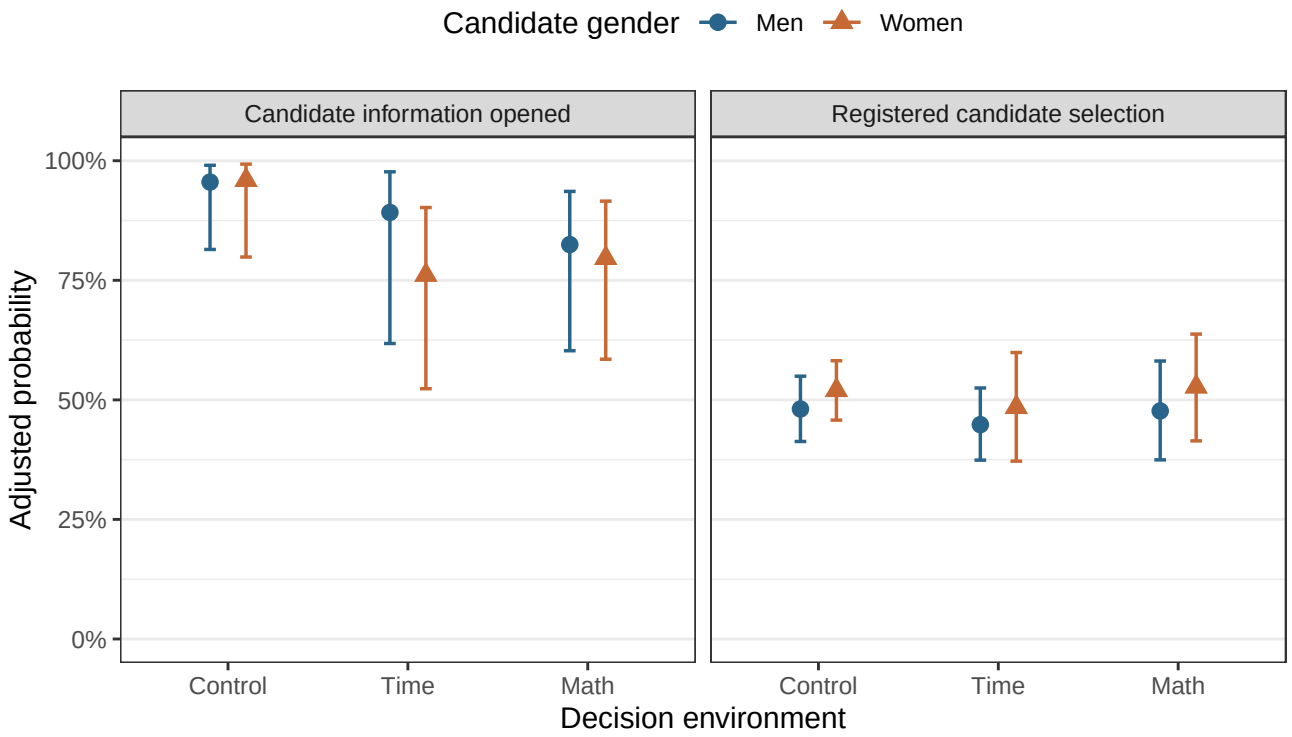


Figure 7: Adjusted information acquisition and registered selection by candidate gender

Notes. Separate mean-bias-reduced logistic models use 750 candidate appearances from 25 eligible completed respondents. Specifications include treatment-by-candidate-gender interactions, respondent age and gender, tertiary education, baseline political interest, task position, and randomized side. Predictions replace only the assigned environment and average over the observed profiles within each candidate-gender group. Bars show approximate pointwise 95% confidence intervals using two-way respondent–candidate clustering and 24 degrees of freedom. Positive-semidefinite covariance correction is used when required. Registered selection is a marginal binary event, not a structural paired-choice model. A permitted task without a selection contributes zero selection events for both profiles while its recorded candidate response remains missing. The profiles' gender and other characteristics remain bundled. Differences between candidate-gender groups are therefore not isolated causal effects of gender, and all predictions condition on study completion.

Table 10: Covariate-adjusted process contrasts relative to the unconstrained condition

Outcome	Contrast	Estimate	95% CI	p
Any description opened	Cost - Baseline	-0.160	[-0.305, -0.015]	0.032
Any description opened	Time - Baseline	-0.143	[-0.318, 0.032]	0.106
Descriptions opened	Cost - Baseline	-0.294	[-0.624, 0.037]	0.079
Descriptions opened	Time - Baseline	-0.254	[-0.634, 0.127]	0.181
Capped first-selection time (s)	Cost - Baseline	1.006	[-0.391, 2.403]	0.150
Capped first-selection time (s)	Time - Baseline	-2.087	[-3.827, -0.348]	0.021
No selection within common window	Cost - Baseline	0.237	[-0.032, 0.506]	0.081
No selection within common window	Time - Baseline	-0.471	[-0.710, -0.232]	<0.001
Both descriptions opened	Cost - Baseline	-0.162	[-0.366, 0.042]	0.114
Both descriptions opened	Time - Baseline	-0.140	[-0.366, 0.085]	0.211

Notes. Contrasts average response-scale differences over the same 25 eligible completed respondents and their 375 analytical tasks. Probability differences are proportions, description differences are counts, and latency differences are seconds. Covariates and working models match Figures 5 and 6. Confidence intervals are respondent-clustered pointwise 95% intervals. Two-sided p-values are unadjusted supplementary tests. The unadjusted primary comparisons and their Holm corrections remain reported separately.

6 Discussion

The experiment examines whether randomized constraints alter political information acquisition, candidate choice, and the scrutiny applied to women candidates. The reported comparisons are restricted to eligible completed interviews, so their causal interpretation depends on assumptions about completion-related selection. Because gender, ideology, party, and facial characteristics remain bundled within observed candidates, gender-indexed comparisons additionally capture heterogeneity across naturally occurring profiles rather than isolated attribute effects. Observed information acquisition locates a stage at which unequal evaluation may emerge but does not, by itself, establish an unconscious psychological mechanism. Combining randomized decision environments, repeated choices, and behavioral measures of political learning connects appearance-based judgment to the acquisition and use of substantive information in the Colombian electoral context.

7 References

Alós-Ferrer, C., & Garagnani, M. (2022). Voting under time pressure. *Judgment and Decision Making, 17*(5), 1072–1093. https://doi.org/10.1017/S1930297500009335

Andersen, D. J., & Ditonto, T. (2020). The importance of candidate sex and partisan preference over time: A multiday study of voter decision making. *The Journal of Politics, 82*(4), 1337–1353. https://doi.org/10.1086/708340

Antonakis, J., & Dalgas, O. (2009). Predicting elections: Child's play! *Science, 323*(5918), 1183. https://doi.org/10.1126/science.1167748

Arel-Bundock, V., Greifer, N., & Heiss, A. (2024). How to interpret statistical models using marginaleffects for R and Python. *Journal of Statistical Software, 111*(9), 1–32. https://doi.org/10.18637/jss.v111.i09

Bahamonde, H., & Sarpila, O. (2024). Physical appearance and elections: An inequality perspective. *Political Psychology, 45*(3), 623–642. https://doi.org/10.1111/pops.12940

Ballew, C. C., II, & Todorov, A. (2007). Predicting political elections from rapid and unreflective face judgments. *Proceedings of the National Academy of Sciences, 104*(46), 17948–17953. https://doi.org/10.1073/pnas.0705435104

Banducci, S. A., Karp, J. A., Thrasher, M., & Rallings, C. (2008). Ballot photographs as cues in low-information elections. *Political Psychology, 29*(6), 903–917.

Bansak, K., Hainmueller, J., Hopkins, D. J., & Yamamoto, T. (2018). The number of choice tasks and survey satisficing in conjoint experiments. *Political Analysis, 26*(1), 112–119. https://doi.org/10.1017/pan.2017.40

Bauer, N. M. (2015). Emotional, sensitive, and unfit for office? Gender stereotype activation and support for female candidates. *Political Psychology, 36*(6), 691–708. https://doi.org/10.1111/pops.12186

Bauer, N. M. (2020). Shifting standards: How voters evaluate the qualifications of female and male candidates. *The Journal of Politics, 82*(1), 1–12. https://doi.org/10.1086/705817

Berggren, N., Jordahl, H., & Poutvaara, P. (2010). The looks of a winner: Beauty and electoral success. *Journal of Public Economics, 94*(1–2), 8–15.

Brysbaert, M. (2019). How many words do we read per minute? A review and meta-analysis of reading rate. *Journal of Memory and Language, 109*, 104047. https://doi.org/10.1016/j.jml.2019.104047

Caballero Germain, G., Carbonell Bellolio, F., López Masle, J., & Villarroel Oyarzún, T. (2021). Medición remota de la velocidad lectora en estudiantes de derecho: Una contribución a la racionalización de la carga académica. *Revista Pedagogía Universitaria y Didáctica del Derecho, 8*(2), 193–212. https://doi.org/10.5354/0719-5885.2021.64735

Card, S. K., Moran, T. P., & Newell, A. (1980). The keystroke-level model for user performance time with interactive systems. *Communications of the ACM, 23*(7), 396–410. https://doi.org/10.1145/358886.358895

Ditonto, T. M., Hamilton, A. J., & Redlawsk, D. P. (2014). Gender stereotypes, information search, and voting behavior in political campaigns. *Political Behavior, 36*(2), 335–358. https://doi.org/10.1007/s11109-013-9232-6

Ditonto, T., & Mattes, K. (2018). Differences in appearance-based trait inferences for male and female political candidates. *Journal of Women, Politics & Policy, 39*(4), 430–450. https://doi.org/10.1080/1554477X.2018.1506206

Hainmueller, J., Hopkins, D. J., & Yamamoto, T. (2014). Causal inference in conjoint analysis: Understanding multidimensional choices via stated preference experiments. *Political Analysis, 22*(1), 1–30. https://doi.org/10.1093/pan/mpt024

Herrmann, M., & Shikano, S. (2016). Attractiveness and facial competence bias face-based inferences of candidate ideology. *Political Psychology, 37*(3), 401–417.

Jenke, L. (2025). An eye tracking study of gender biased information acquisition in candidate evaluation. *Political Psychology, 46*(3), 673–690. https://doi.org/10.1111/pops.13031

Kocher, M. G., Schindler, D., Trautmann, S. T., & Xu, Y. (2019). Risk, time pressure, and selection effects. *Experimental Economics, 22*(1), 216–246. https://doi.org/10.1007/s10683-018-9576-1

Kosmidis, I., & Firth, D. (2021). Jeffreys-prior penalty, finiteness and shrinkage in binomial-response generalized linear models. *Biometrika, 108*(1), 71–82. https://doi.org/10.1093/biomet/asaa052

Lawson, C., Lenz, G. S., Baker, A., & Myers, M. (2010). Looking like a winner: Candidate appearance and electoral success in new democracies. *World Politics, 62*(4), 561–593.

Lenz, G. S., & Lawson, C. (2011). Looking the part: Television leads less informed citizens to vote based on candidates'

appearance. *American Journal of Political Science, 55*(3), 574–589.

Lodge, M., McGraw, K. M., & Stroh, P. (1989). An impression-driven model of candidate evaluation. *American Political Science Review, 83*(2), 399–419. https://doi.org/10.2307/1962397

MacKenzie, I. S. (1992). Fitts' law as a research and design tool in human-computer interaction. *Human–Computer Interaction, 7*(1), 91–139. https://doi.org/10.1207/s15327051hci0701_3

Marín, A., Bahamonde, H., & Canales, A. (2026). *Faces or Facts? Experimental Evidence on Time Pressure, Information Costs, and Gender Bias in Electoral Choice.* AEA RCT Registry, AEARCTR-0017574. https://doi.org/10.1257/rct.17574-1.3

Mattes, K., Spezio, M., Kim, H., Todorov, A., Adolphs, R., & Alvarez, R. M. (2010). Predicting election outcomes from positive and negative trait assessments of candidate images. *Political Psychology, 31*(1), 41–58.

Olivola, C. Y., Tingley, D., & Todorov, A. (2018). Republican voters prefer candidates who have conservative-looking faces: New evidence from exit polls. *Political Psychology, 39*(5), 1157–1171. https://doi.org/10.1111/pops.12489

Ordóñez, L., & Benson, L., III. (1997). Decisions under time pressure: How time constraint affects risky decision making. *Organizational Behavior and Human Decision Processes, 71*(2), 121–140. https://doi.org/10.1006/obhd.1997.2717

Payne, J. W., Bettman, J. R., & Johnson, E. J. (1988). Adaptive strategy selection in decision making. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 14*(3), 534–552. https://doi.org/10.1037/0278-7393.14.3.534

Payne, J. W., Bettman, J. R., & Luce, M. F. (1996). When time is money: Decision behavior under opportunity-cost time pressure. *Organizational Behavior and Human Decision Processes, 66*(2), 131–152. https://doi.org/10.1006/obhd.1996.0044

Rieskamp, J., & Hoffrage, U. (2008). Inferences under time pressure: How opportunity costs affect strategy selection. *Acta Psychologica, 127*(2), 258–276. https://doi.org/10.1016/j.actpsy.2007.05.004

Rohrbach, T., & Schönhagen, P. (2025). Gender on the mind? Gender heuristics and rationalizations in candidate evaluations. *Political Psychology, 46*(4), 766–784. https://doi.org/10.1111/pops.13036

Rosar, U., Klein, M., & Beckers, T. (2008). The frog pond beauty contest: Physical attractiveness and electoral success of the constituency candidates at the North Rhine-Westphalia state election of 2005. *European Journal of Political Research, 47*(1), 64–79.

Sauro, J. (2009). Estimating productivity: Composite operators for keystroke level modeling. In J. A. Jacko (Ed.), *Human-Computer Interaction. New Trends* (Lecture Notes in Computer Science, Vol. 5610, pp. 352–361). Springer. https://doi.org/10.1007/978-3-642-02574-7_40

Todorov, A., Mandisodza, A. N., Goren, A., & Hall, C. C. (2005). Inferences of competence from faces predict election outcomes. *Science, 308*(5728), 1623–1626.

A Task duration by study version

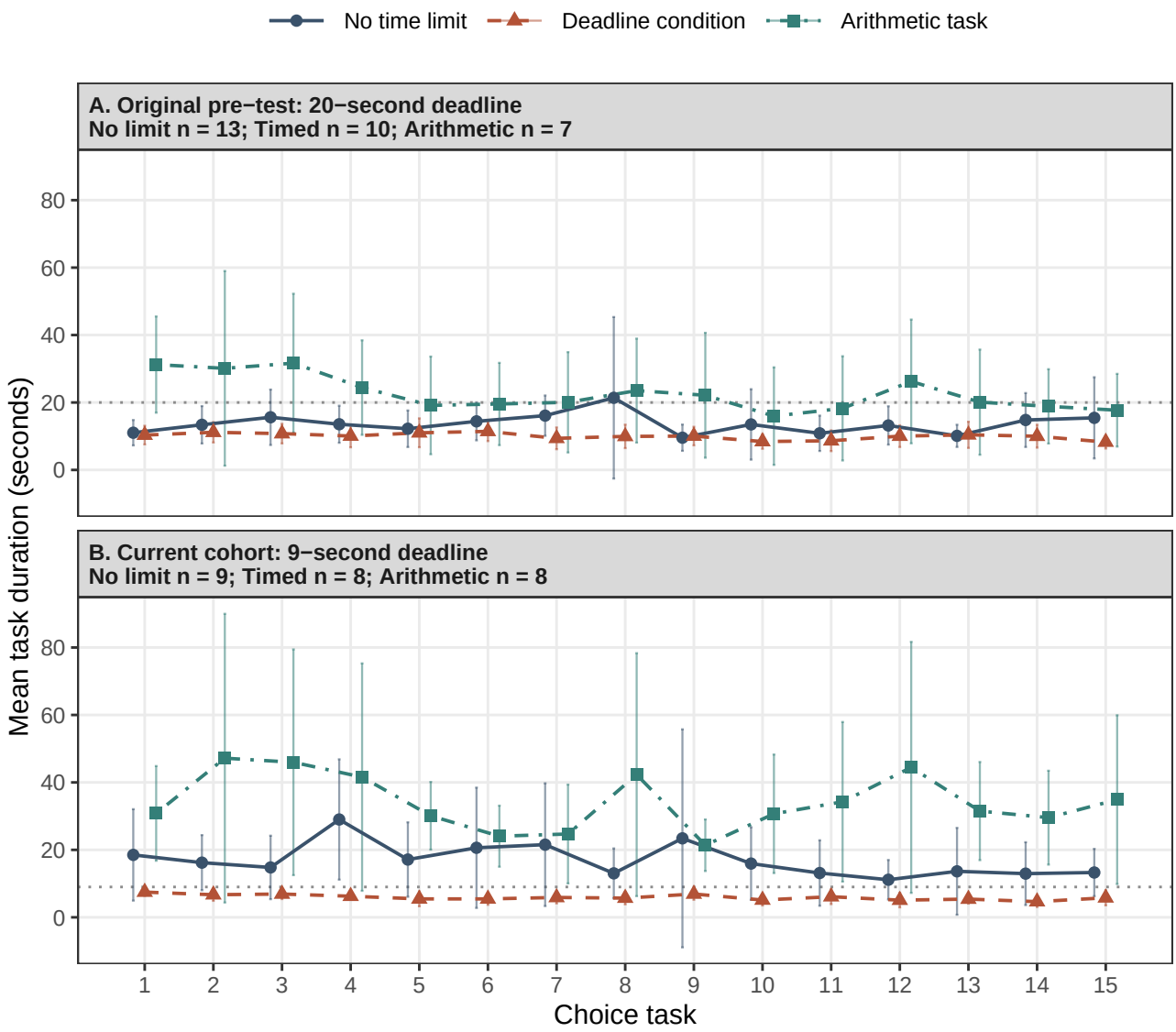


Figure A1: Mean choice-task duration by condition and study version among eligible completes

Notes. Panel A comprises the original 30 eligible pre-test participants who reached the final study screen, submitted all 15 analytical tasks, and recorded at least 13 valid candidate choices. The unconstrained, exploratory timed, and arithmetic-cost conditions include 13, 10, and 7 participants, respectively. Each contributes one observation to each of the 15 analytical choice tasks, for 450 task observations. The 5 practice tasks are excluded. Points report arithmetic means, and bars show pointwise 95% Student-t confidence intervals across respondents within each condition and task. Time runs from the display of both loaded photographs until manual or automatic task submission. It includes information acquisition and arithmetic operations when applicable, but excludes image loading and follow-up questions. Up to 2 timed submissions without a candidate selection are permitted per retained interview and remain in the figure. No long durations are removed or capped. Panel A's dotted line marks the exploratory 20-second deadline. Confidence limits may extend below zero, although observed durations are nonnegative. Comparisons are descriptive and conditional on study completion, which may itself be affected by treatment. Panel B includes 25 qualifying completes from the revised 9-second version, contributing 375 analytical task observations. Its condition-specific respondent counts appear in the panel header. Recorded navigation violations and terminated cases are excluded from this panel but retained in the source data. Both panels use the same seconds scale. Their dotted lines mark their respective deadlines. The two cohorts are distinct, and differences between panels are not randomized effects of deadline length.

B Candidate characteristics and registered selection

Table B1: Displayed and selected candidate characteristics by condition

Attribute	Level	Condition	Shown	Chosen	Chosen / shown	Among chosen
Age category	Older	Baseline	139	69	49.6	51.1
Age category	Older	Time	130	70	53.8	62.5
Age category	Older	Cost	123	65	52.8	54.2
Age category	Younger	Baseline	131	66	50.4	48.9
Age category	Younger	Time	110	42	38.2	37.5
Age category	Younger	Cost	117	55	47.0	45.8
Dominance	High	Baseline	139	67	48.2	49.6
Dominance	High	Time	126	65	51.6	58.0
Dominance	High	Cost	126	61	48.4	50.8
Dominance	Low	Baseline	131	68	51.9	50.4
Dominance	Low	Time	114	47	41.2	42.0
Dominance	Low	Cost	114	59	51.8	49.2
Gender	Men	Baseline	143	69	48.3	51.1
Gender	Men	Time	121	54	44.6	48.2
Gender	Men	Cost	122	58	47.5	48.3
Gender	Women	Baseline	127	66	52.0	48.9
Gender	Women	Time	119	58	48.7	51.8
Gender	Women	Cost	118	62	52.5	51.7
Ideology	Center	Baseline	88	46	52.3	34.1
Ideology	Center	Time	76	36	47.4	32.1
Ideology	Center	Cost	79	38	48.1	31.7
Ideology	Left	Baseline	97	55	56.7	40.7
Ideology	Left	Time	91	52	57.1	46.4
Ideology	Left	Cost	82	57	69.5	47.5
Ideology	Right	Baseline	85	34	40.0	25.2
Ideology	Right	Time	73	24	32.9	21.4
Ideology	Right	Cost	79	25	31.6	20.8

Notes. The table includes 750 candidate appearances from 71 distinct candidates evaluated by 25 eligible completed respondents. Each submitted analytical task contributes its 2 displayed candidate profiles. Shown counts profile appearances, and chosen counts registered selections. Chosen / shown is the percentage of displayed profiles with that characteristic that were selected. Among chosen is the percentage of registered selections with that characteristic, calculated separately for each attribute and condition. The latter percentages therefore sum to 100 within each attribute and condition. A retained task without a selection contributes 2 shown profiles and no chosen profile. The original missing choice is not imputed. Ideology refers to the classification in the candidate's available description. The age grouping uses the supplied younger/older profile classification rather than age in years. The cost condition requires an arithmetic operation to access information.

Table B2: Adjusted associations between candidate characteristics and selection

Attribute	Contrast	Condition	Estimate	95% CI	p
Gender	Women - Men	Baseline	4.9	[-8.2, 18.0]	0.4447
Gender	Women - Men	Time	3.9	[-10.6, 18.4]	0.5833
Gender	Women - Men	Cost	4.9	[-15.0, 24.9]	0.6134
Ideology	Left - Center	Baseline	5.6	[-21.4, 32.5]	0.6743
Ideology	Right - Center	Baseline	-12.2	[-38.2, 13.9]	0.3442
Ideology	Left - Center	Time	11.0	[-6.2, 28.1]	0.1983
Ideology	Right - Center	Time	-13.7	[-31.9, 4.5]	0.1322
Ideology	Left - Center	Cost	21.3	[2.4, 40.3]	0.0288
Ideology	Right - Center	Cost	-16.4	[-35.6, 2.9]	0.0916
Dominance	High - Low	Baseline	-3.8	[-18.8, 11.2]	0.6078
Dominance	High - Low	Time	11.3	[3.1, 19.6]	0.0091
Dominance	High - Low	Cost	-2.3	[-14.9, 10.3]	0.7116
Age category	Older - Younger	Baseline	-1.8	[-20.2, 16.6]	0.8410
Age category	Older - Younger	Time	14.0	[0.8, 27.2]	0.0387
Age category	Older - Younger	Cost	5.1	[-11.8, 22.0]	0.5391

Notes. Estimates are within-condition differences in the probability of a registered candidate selection, expressed in percentage points. The joint linear probability model includes candidate gender, displayed ideology, dominance classification, and candidate age group, together with their interactions with condition, randomized side, and task-order indicators. Reference categories are men, centrist ideology, low dominance, and the younger age category. Confidence intervals are pointwise 95% intervals with two-way clustering by respondent and candidate. Two-sided p-values are unadjusted exploratory tests and do not enter the primary multiplicity families. When required, the software's standard positive-semidefinite covariance correction is applied. The model uses 750 candidate appearances from 25 eligible completed interviews. Retained submissions without a choice contribute zero registered selections for both displayed candidates, while the original candidate response remains missing. Because candidate attributes remain bundled within real profiles, these contrasts are adjusted associations, not attribute-level average marginal component effects. They do not isolate the effect of reading political information.

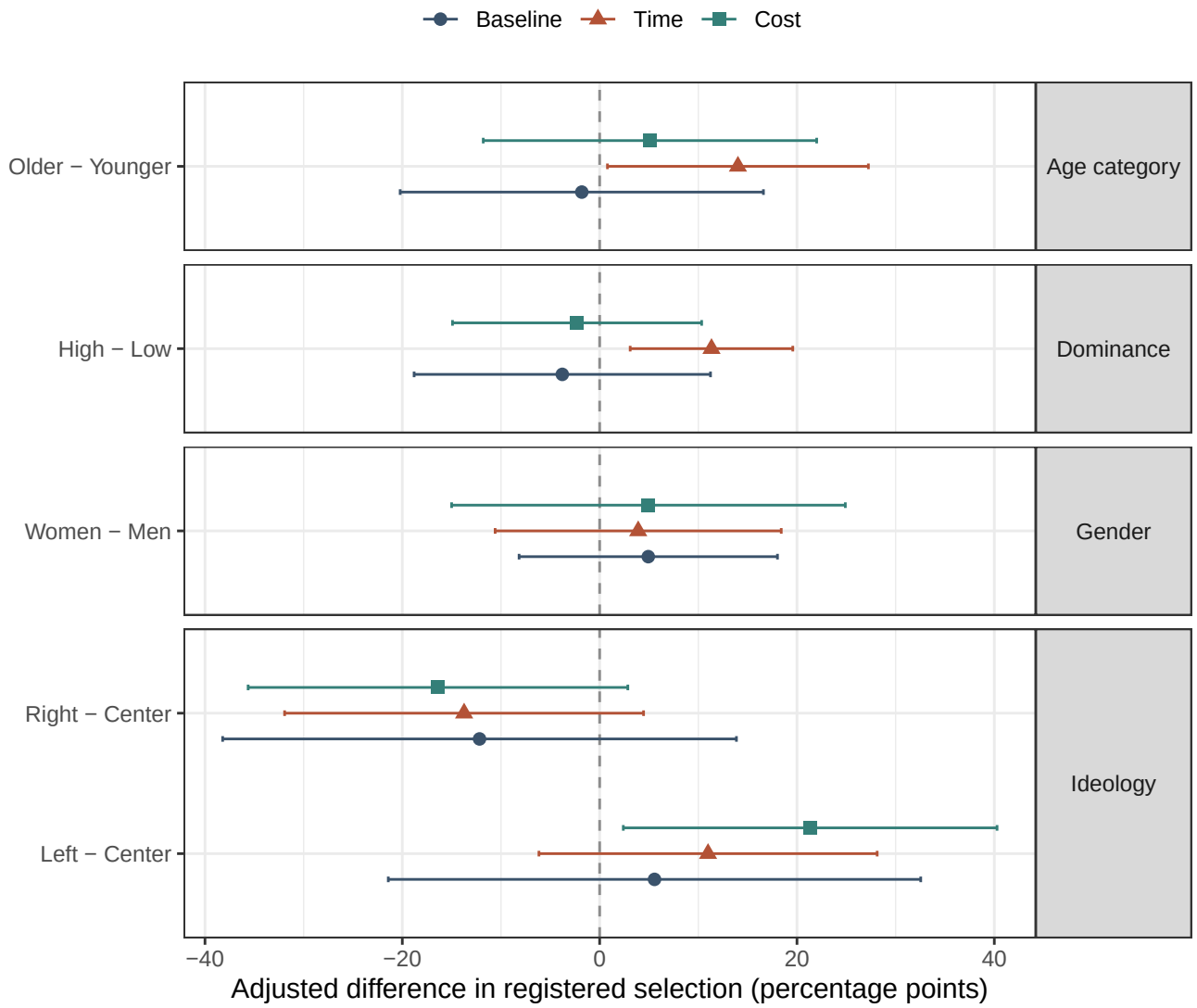


Figure B1: Candidate characteristics and registered selection by decision environment

Notes. Points report adjusted within-condition differences in registered selection in percentage points from the joint model in Table B2. Bars show two-way clustered pointwise 95% confidence intervals. The vertical line marks no difference from the relevant reference category. Estimates use 750 candidate appearances from 25 eligible completed interviews. The layout follows a conjoint coefficient plot, but the estimates are associations across naturally occurring candidate profiles rather than causal effects of independently randomized attributes. Baseline has no time or arithmetic constraint, time imposes a 9-second deadline, and cost requires an arithmetic operation.

C Information acquisition by respondent and candidate gender

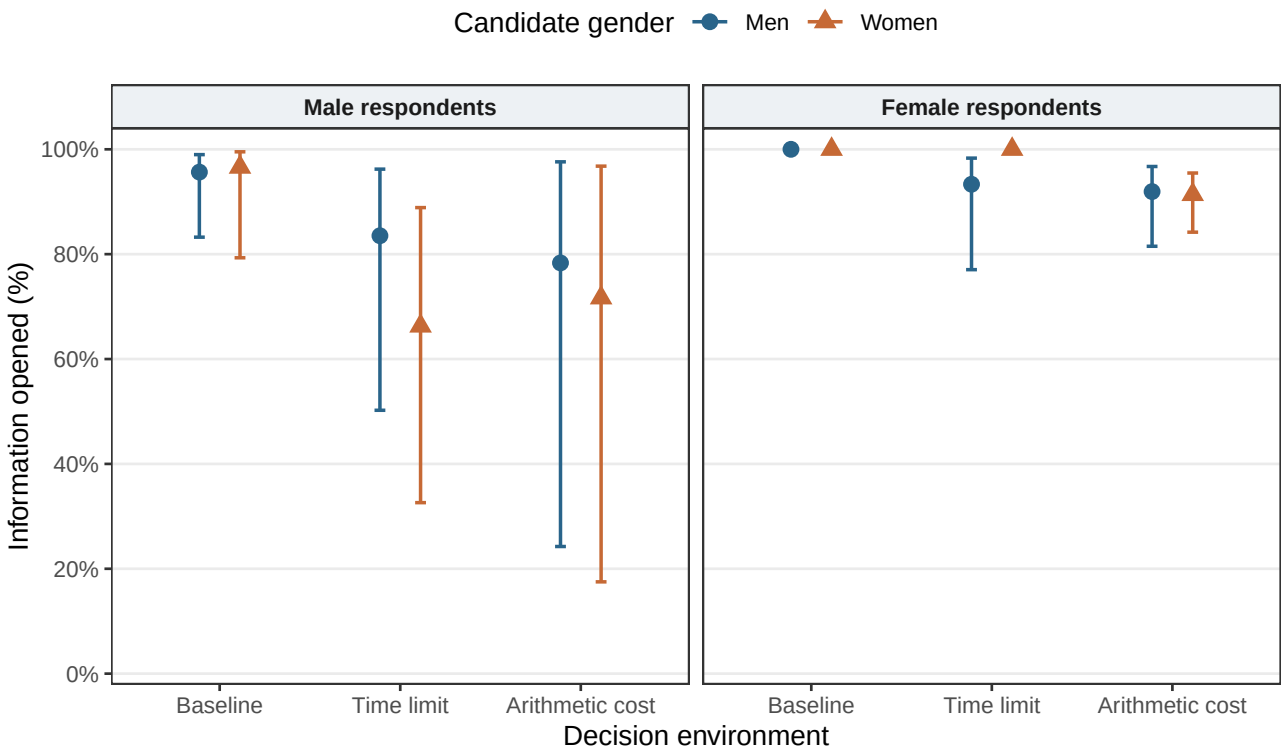


Figure C1: Candidate information acquisition among male and female respondents

Notes. Panels distinguish male and female respondents. Within each panel, points show the probability that information about a male or female candidate was opened at least once in each experimental condition. The points equal the observed opening rates, not differences between respondent groups. Bars show pointwise 95% confidence intervals from a saturated binomial-logit model of condition, candidate gender, respondent gender, and their interactions. The covariance is clustered by respondent and candidate, with a positive-semidefinite correction when required. Student-t critical values use 24 degrees of freedom. Intervals are calculated on the log-odds scale and transformed to probabilities. The figure uses 750 candidate appearances from 25 eligible completed interviews. Respondent counts by condition are Baseline men 6 and women 3. Time men 6 and women 2. Cost men 4 and women 4. Participants reporting another gender identity or not reporting gender are outside this specific comparison, but remain in the other analyses if eligible. This excludes 0 completed respondents from the present figure. Retained tasks without a candidate selection still contribute their recorded information-opening outcomes. The measure does not establish that the text was read or count repeated clicks. These exploratory, completion-conditioned associations are not causal effects of respondent or candidate gender.

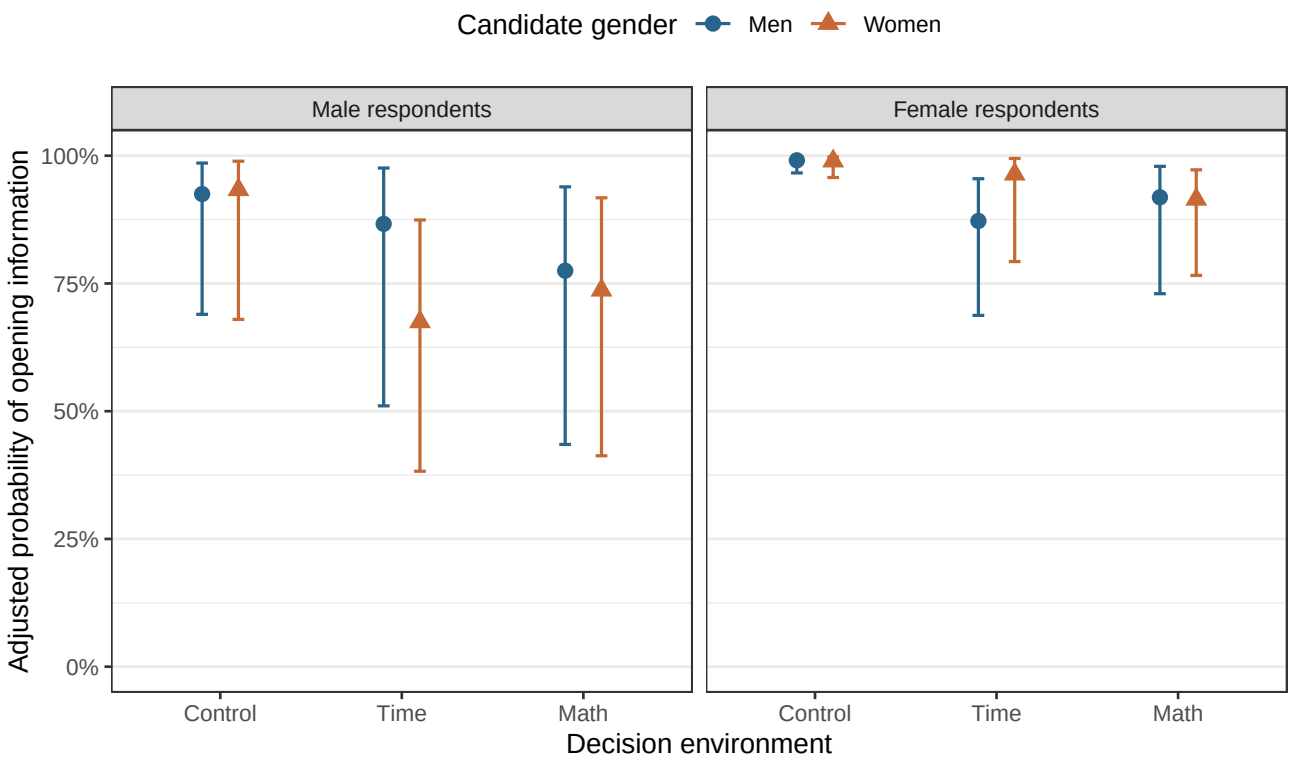


Figure C2: Covariate-adjusted information acquisition by respondent and candidate gender

Notes. Panels distinguish observed male and female respondents. Within each panel, points show average predicted information-opening probabilities for profiles of male and female candidates in each decision environment. A mean-bias-reduced logistic model includes the three-way interaction among treatment, candidate gender, and respondent gender, adjusting for age, tertiary education, baseline political interest, task position, and randomized side. The model uses 750 candidate appearances from 25 eligible completed respondents. Respondent counts by condition are Baseline men 6 and women 3. Time men 6 and women 2. Cost men 4 and women 4. Each treatment prediction averages over the observed members and profiles of the relevant respondent–candidate-gender subgroup, without changing either gender. Bars are approximate pointwise 95% confidence intervals based on two-way respondent–candidate clustering, 24 degrees of freedom, and logit transformation of the average probability. Positive-semidefinite covariance correction is used when required. Subgroup differences are exploratory, completion-conditioned associations, not causal effects of respondent or candidate gender. Information opening does not establish that the text was read.